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Rescheduling Part 2 of the 11+ reduces injury burden and increases compliance in semi-professional football

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Although the 11+ program has been shown to reduce injuries in sub-elite football, program compliance is typically poor, suggesting that strategies to optimize delivery are necessary. This study investigated the effect of rescheduling Part 2 of the three-part 11+ program on program effectiveness. Twenty-five semi-professional football clubs were randomly allocated to either a Standard-11+ (n = 398 players) or $P2_{post}$ group (n = 408 players). Both groups performed the 11+ program at least twice a week throughout the 2017 football season. The Standard-11+ group performed the entire 11+ program before training activities commenced, whereas the $P2_{post}$ group performed Parts 1 and 3 of the 11+ program before and Part 2 after training. Injuries, exposure, and individual player 11+ dose were monitored throughout the season. No significant between group difference in injury incidence rate ($P2_{post}$ vs Standard-11+ = 11.8 vs 12.3 injuries/1000 h) was observed. Severe time loss injuries > 28 days (33 vs 58 injuries; $P < .002$) and total days lost to injury (4303 vs 5815 days; $P < .001$) were lower in the $P2_{post}$ group. A higher 11+ program dose was observed in the $P2_{post}$ (29.1 doses; 95% CI 27.9-30.1) versus Standard-11+ group (18.9 doses; 95% CI 17.6−20.2; $P < .001$). In semi-professional football, rescheduling Part 2 of the 11+ program to the end of training maintained the effectiveness of the original 11+ program to reduce injury incidence. Importantly, rescheduling Part 2 improved player compliance and reduced the number of severe injuries and total injury burden, thereby enhancing effectiveness of the 11+ program.

KEYWORDS
11+, FIFA 11+, football, injury prevention, soccer

1 | INTRODUCTION

Football is a contact sport characterized by periods of high-intensity activity, which carries an inherent risk of injury.[1-3] While participation in football is associated with improved health,[4] injury is often the reason players discontinue participating in the sport, leading to longer term disability and substantial medical costs.[5] The high injury incidence and burden observed in sub-elite football[6] suggest there is a need to develop injury prevention strategies. Importantly, injury prevention strategies may not only help maintain long-term player participation but also reduce health costs associated with sporting injury,[5] a finding that has been specifically observed following implementation of the 11+ program in amateur football.[7,8]

The 11+ program was developed and disseminated by Fédération Internationale de Football Association (FIFA) to reduce football injuries. The program consists of three parts; Parts 1 and 3 focus on running-based activity including dynamic actions and accelerations, whereas Part 2 focuses on strengthening and neuromuscular control exercises. The 11+ program was designed to be delivered as a 20- to 25-minute

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"warm-up" before commencing other training activities, without the need for specialized expertise or equipment. This is important for sub-elite sports where resources are typically scarce with limited access to staff[6,9] and, as a result, injury prevention programs are usually coach-led.[10] Previous research has shown that injury rates can be reduced by ≥ 40% when players complete the 11+ program at least twice per week,[11] with higher program compliance and dose exposure associated with increased program effectiveness.[12] However, despite the proven effectiveness of the 11+ program, only 10% of FIFA's member associations have endorsed the program and several studies highlight low 11+ program compliance.[13-15]

Poor adoption and compliance rates of the 11+ program have been explained by (a) the time required to complete and boredom associated with the program,[10,16] (b) fatigue and soreness caused by exercises contained in Part 2,[10,15-17] and (c) a lack of awareness and knowledge of how to perform the program.[16,18] Furthermore, given the importance of strengthening exercises in reducing injury risk,[19] it is concerning that research has shown that the strengthening exercises performed in Part 2 of the 11+ program are often modified or not performed.[15] Potential fatigue caused by the exercises in Part 2 may contribute to why compliance to the full 11+ program is poor,[15] with fatigue considered by practitioners to be a primary injury risk factor in football.[20,21] Interestingly, it has been found that performing an exercise in Part 2, the Nordic Hamstring Exercise (NHE), prior to football activity exacerbates eccentric hamstring fatigue,[22,23] although administering these exercises after training did not affect the exercise stimulus. Furthermore, rescheduling the NHE to the end of training not only maintained the effectiveness of the NHE in terms of improving eccentric hamstring strength[24] and reducing hamstring muscle injury incidence,[22] but was also associated with enhanced compliance to the intervention.[24,25] These findings from one exercise may suggest rescheduling Part 2 of the 11+ *after* training as a practical alternative to address identified barriers to implementation, such as program duration, exercise difficulty/fatigue, and boredom.

Exploring ways to ensure that all three components of the 11+ program (Parts 1, 2, and 3) are performed, including the strengthening exercises in Part 2, could therefore improve program effectiveness, although research is required to confirm or refute this notion. Therefore, the aim of this study was to determine whether rescheduling the 11+ program, such that Parts 1 and 3 are performed at the beginning, and Part 2 at the end of training, affected program effectiveness, compared with the Standard 11+ program performed in its entirety at the beginning of training.

2 | MATERIALS AND METHODS

2.1 | Participant recruitment

Twenty-five sub-elite football clubs, each comprising 2-3 teams, volunteered to participate in the study during the 2017 season. The clubs consisted of 4 Tier 2 (National Premier League) and 21 Tier 3 (Regional League) clubs in which all players received payment to play. Club and player recruitment and engagement were performed according to the Sports Setting Matrix,[26] which was developed to help identify key stakeholders and "levels" of engagement required for successful implementation of an injury prevention program.[26] In this study, this involved gaining approval from

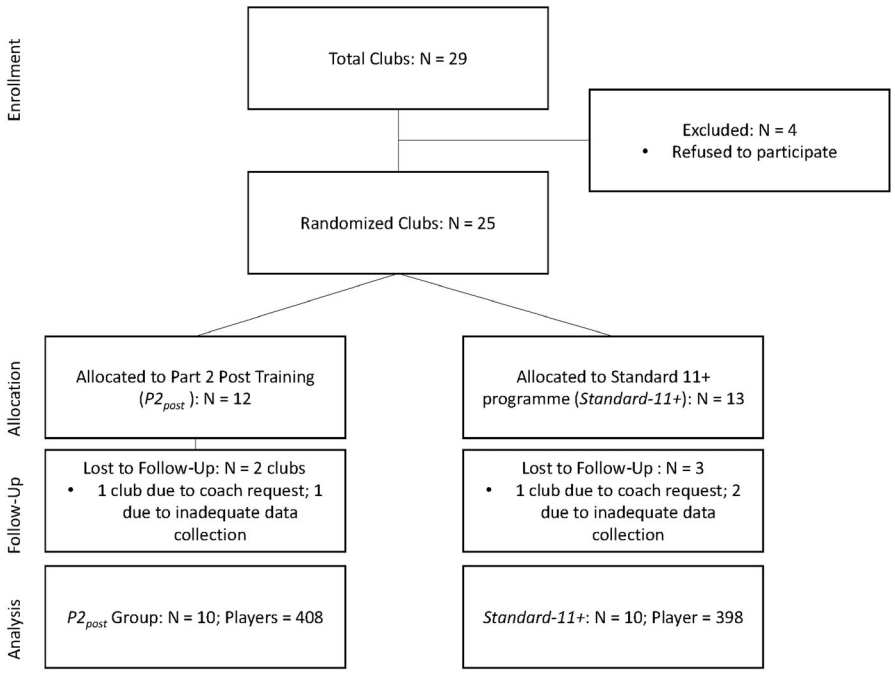

FIGURE 1 Consolidated standards of reporting trials (CONSORT) diagram of the flow of participants in the study

the National and State Federations, engaging with regional associations and presenting to club officials, coaching staff, and players about the study. Before data collection, all players provided signed informed consent. All procedures were approved by the University of Wollongong Human Research Ethics Committee (15/340).

Clubs were randomly allocated to either: (a) Standard-11+ or (b) a $P2_{post}$ groups. All groups were instructed to complete the full 11+ program a minimum of twice per week at training, and Parts 1 and 3 before matches (Figure 1). The Standard-11+ group performed all three parts of the 11+ program at the start of training as a warm-up, whereas the $P2_{post}$ group were instructed to perform Parts 1 and 3 at the start of training as a warm-up and Part 2 of the 11+ program at the end of training during the cool down period. Coaches and players were permitted to include additional exercises, including those involving a ball, into the warm-up and cool down once the 11+ exercises were completed.

2.2 | Training to implement the 11+ program

Before implementing the 11+ program, the chief investigator (MW) presented information to coaches, club officials, and medical staff about (a) the rate and burden of injury in sub-elite football, (b) the 11+ program and its effectiveness, (c) barriers affecting uptake of the 11+ program, (d) coach education regarding the importance of their role in 11+ program adoption,[8,27] (v) when to progress the 11+ exercises, and (e) the role of the primary data collector (PDC) in program delivery. Coaches and medical staff were also shown videos of all exercises in the 11+ program, given explanations for the purpose and required technique for each exercise in the program, instructed on the process and criteria for stage progress in Part 2, and informed of the positive impact coach delivery has upon effectiveness of the program.[28] The 11+ program instructions allow players to progress through the three stages of Part 2 as they felt comfortable, on the grounds that the coach and/or PDC were satisfied with their exercise technique. However, we applied progression restrictions in accordance with previous research,[29] whereby players remained at Level 1 of Part 2 for a minimum of 2 weeks initially and progressed to Level 3 after a minimum of 6 weeks. In the event a player missed a week of training due to injury, they were required to return to a lower level of Part 2 for a minimum of 1 week. At the information sessions for club officials and coaches, it was made clear that the PDC would be trained in implementing the 11+ program and would attend training sessions to coordinate the program. Coaches were not required to deliver the program, but were encouraged to support its implementation. Paper and digital copies of the 11+ poster and field set up cards were also provided.

2.3 | Program compliance and injury data collection

To determine the effectiveness of rescheduling Part 2 of the 11+ program, individual player exposure to the 11+ program was monitored. Given the cluster randomized controlled experimental design adopted, in which players were instructed to complete a specific set of exercises, the term "compliance" was used to assess player dose exposure and was used as a continuous predictor for analysis.[30] A player was deemed to have performed the 11+ only when they completed all components of the program during that session. Data pertaining to program compliance and player injuries for each participating team were collected by an allocated onsite primary data collector (PDC), a qualified sports trainer, who attended all training sessions and matches.[31,32] All PDCs completed at least 6 hours of training including how to record program compliance, injury and exposure data recording, injury definitions, and details regarding the correct delivery of the 11+ program (see Appendix S1 for operational definitions).[6] Training included scenario-based examples, and the primary researcher (MW) was in weekly contact with PDCs during the season to review data collected. A time loss injury was defined as an "injury that results in a player being unable to fully participate in matches or training."[31] Players were deemed to have recovered from injury once they had returned to full training/match participation or were considered eligible for team selection.[31] Injury records were obtained during all training sessions (2-3 per week) and matches including preseason, in-season, and finals (28-34 weeks).

The primary outcome variables for this study included program compliance, represented by the total 11+ dose per player, and program effectiveness in reducing injury. Program effectiveness was represented by injury count (total number of injuries), injury severity, injury incidence (total number of injuries/total football exposure (h) × 1000 h), the number of days lost to injury, and injury type, locations, and mechanisms. Injury burden was then calculated (injury incidence × mean absence (days) per injury).[33]

2.4 | Statistical analysis

To account for the potential club cluster effect on outcomes, a generalized estimated equation (GEE) was performed with Poisson distribution used to assess between group differences in injury count, injury severity, and days lost to injury, with participant group, 11+ dose, and total soccer exposure (h) entered as predictor variables. A second GEE was performed to determine significant differences between the two groups (Standard-11+ and $P2_{post}$) for 11+ dose exposure with participant group and exposure imputed as predictor variables. A Mann-Whitney U test was used to assess differences between 11+ doses for each group (SPSS v25, IBM).

TABLE 1 Player exposure, compliance, injury count, and severity for participants in the Standard-11+ and $P2_{post}$ group

Variable	Standard-11+ Group (n = 398)	$P2_{post}$ Group (n = 408)	*P* value
Player characteristics & positions	(mean [95% CI])	(mean [95% CI])	
Age (years)	24.8 [24.0,25.6]	23.8 [23.0,24.7]	.218
Height (cm)	176.9 [176.2,177.6]	178.3 [177.5,179.1]	.061
Weight (Kg)	79.3 [78.9,79.7]	78.3 [77.9,78.7]	.120
Goalkeepers (%)	10.1	9.9	
Defenders (%)	32.2	32.3	
Midfielders (%)	32.7	31.2	
Strikers (%)	25.0	26.6	
Player exposure & compliance	(mean [95% CI])	(mean [95% CI])	
Total football exposure (h)	26 062.1	28 541.4	.972
Total training sessions (n)	51.6 [50.3,52.9]	49.7 [48.6, 50.9]	.721
Total 11 + player doses (n)	7625	11 871	.004*
Total 11 + player dose (sessions)	18.9 [17.6, 20.2]	29.1 [27.9, 30.1]	<.001*
11+ player dose/training Session (%)	32.7 [31.1, 34.3]	57.7 [56.2, 59.2]	-
11+ Player dose/Exposure (h)	0.27 [0.26, 0.28]	0.42 [0.41, 0.43]	-
Injury count	n (% total)	n (% total)	
Total injuries (n)	320 (48.7)	337 (51.3)	.825
Days lost to injury (n)	5815*	4303*	.026*
Injury severity	n (% total)	n (% total)	
Minimal: 1-3 days lost	60 (19)	95 (28)	.335
Mild: 4-7 days lost	93 (29)	104 (31)	.832
Moderate: 8-28 days lost	108 (34)	105 (31)	.881
Severe: >28 days lost	59 (18)	33 (10)	.012*

*Significant difference between the Standard-11+ and $P2_{post}$ groups.

Injury incidence rate (IIR) ratios (±95% confidence intervals [CI]) were calculated to compare injury locations and types between the groups. In addition, IIRs were also determined for a subset of players (Standard-11+, n = 185 and $P2_{post}$, n = 226) who participated in a previous surveillance season (2016), in which no club implemented the 11+ or any other known injury prevention program.[6] Therefore, the data collected in 2016, which was conducted by the same research team and collection procedures, served as baseline, pre-intervention data pertaining to injury incidence. All IIR ratio analysis was performed via Hopkin's "compare and combine" analysis to determine clinical inference ranging from "most unlikely to be beneficial < 0.5%" to "most likely to be beneficial > 99.5%."[34]

3 | RESULTS

A total of 806 male players consented to participate in this study with 398 players in the Standard-11+ group and 408 players in $P2_{post}$ (Table 1). Player football exposure, compliance and injury incidence, and severity for participants in the Standard-11+ and $P2_{post}$ groups are presented in Table 1. A total of 657 time loss injuries were recorded during 54 604 hours of football exposure (training and matches combined) across both groups. A similar number of time loss injuries were observed in both groups. However, significantly higher 11+ dose, a lower number of severe injuries and correspondingly lower total number of days lost to injury were observed in the $P2_{post}$ group compared with the Standard-11+ group (Table 1).

The injury location differed between participant groups, with a significantly lower incidence of ankle and recurrent injuries in the $P2_{post}$ group, whereas the incidence of quadriceps muscle and contusion injury was lower in the Standard-11+ group (Table 2). Total injury burden was also lower in the $P2_{post}$ group, with lower time lost (days) associated with non-contact, recurrent, and hamstring muscle injuries compared with the Standard-11+ group (Tables 3 and 4). There was also a significantly lower incidence of non-contact ankle injury injuries in $P2_{post}$ (Table 4).

Comparing the subset of the current data (Total combined IIR = 12.5 injuries/1000 h; Standard-11+ IIR = 12.3 injuries/1000 h, total injuries = 206; $P2_{post}$ group IIR = 12.9

TABLE 2 Injury pattern, incidence rate, and rate ratios for participants in the Standard-11+ and $P2_{post}$ Groups

	Standard-11+ Group		$P2_{post}$ Group				
	No. of Injuries	IR per 1000 h [95% CI]	No. of Injuries	IR per 1000 h [95% CI]	IRR [95% CI][a]	*P* value (No. of Injuries)	Clinical Inference IRR[b]
Injury location							
Thigh	68	2.6 [2.4,2.7]	79	2.8 [2.5, 3.0]	1.1 [0.8, 1.5]	.806	Unclear
Hamstrings	51	2.0 [1.8,2.2]	45	1.6 [1.3,1.8]	0.8 [0.5, 1.2]	.370	Possibly beneficial
Quadriceps	17	0.7 [0.5,0.9]	34	1.2 [0.9,1.4]	1.8 [1.0, 3.7]	.098	Possibly harmful
Knee	59	2.3 [2.1,2.5]	60	2.1 [1.9, 2.4]	0.9 [0.7, 1.3]	.857	Likely trivial effect
Ankle	55	2.1 [1.9,2.4]	39	1.4 [1.1,1.6]	0.7 [0.4, 0.9]	.132	Possibly beneficial
Hip/groin	41	1.6 [1.4,1.7]	50	1.8 [1.6,2.1]	1.1 [0.7, 1.7]	.618	Likely trivial effect
Lower leg/Achilles tendon	40	1.5 [1.1,1.9]	47	1.6 [1.1,2.1]	1.1 [0.7, 1.6]	.745	Possibly trivial effect
Injury type							
Muscle injury/strain	135	5.2 [4.7,5.4]	139	4.9 [4.4,5.3]	0.9 [0.7, 1.2]	.882	Very likely trivial
Sprain/ligament injury	85	3.3 [3.0,3.7]	77	2.7 [2.5,3.0]	0.8 [0.6 to 1.1]	.812	Possibly beneficial
Hematoma/contusion	38	1.5 [1.2,1.8]	64	2.2 [1.9,2.5]	1.5 [1.0 to 2.3]	.116	Likely harmful
Fracture	13	0.5 [0.2,0.8]	10	0.4 [0.2,0.9]	0.7 [0.3, 1.6]	.832	Unclear
Other bone injury	10	0.4 [0.1,0.9]	9	0.3 [0.1,0.6]	1.0 [0.4, 2.5]	.891	Unclear
Meniscus/cartilage	10	0.4 [0.1,0.8]	7	0.2 [0.08,0.5]	0.6 [0.2, 1.7]	.765	Unclear
Tendon injury	9	0.3 [0.1,0.7]	12	0.4 [0.1,0.8]	1.2 [0.5, 2.9]	.832	Unclear
Injury mechanism							
Non-contact	189	7.3 [7.0,7.5]	177	6.2 [5.8,6.6]	0.9 [0.7, 1.1]	.367	Possibly beneficial
Contact	132	5.1 [4.7,5.5]	161	5.6 [5.2,5.9]	1.1 [0.9, 1.4]	.332	Very likely trivial
Recurrent	67	2.6 [2.1,2.9]	48	1.7 [1.4,2.1]	0.7 [0.5, 0.9]	.242	Likely beneficial
Total injuries	320	12.3 [10.4, 14.1]	337	11.8 [10.1, 13.9]	1.0 [0.8 to 1.1]	.579	Likely trivial

[a]IRR—incidence rate ratio for injury incidence ($P2_{post}$ group: Standard-11+ group).
[b]Clinical inference of $P2_{post}$ is more beneficial/harmful than the Standard-11+ determined from IRR via Hopkins "combine and compare" analysis.[34]

TABLE 3 Injury burden and days lost to injury for participants in Standard-11+ and $P2_{post}$ Groups

	Standard-11+ group		$P2_{post}$ group		
	Days lost per 1000 h	Total days lost	Days lost per 1000 h	Total days lost	*P* value (Total days lost)
Injury location					
Thigh	45.7	1133	28.5	843	.058
Hamstrings	39.3	1001	19.6	559	.006*
Quadriceps	6.2	162	8.9	254	.464
Knee	60.7	1622	41.7	1190	.203
Ankle	30.4	797	17.2	490	.300
Hip/groin	20.8	535	19.2	547	.867
Lower leg/Achilles tendon	26.4	697	13.6	371	.127
Injury type					
Muscle injury/strain	75.6	1970	48.7	1390	.080
Sprain/ligament injury	66.0	1720	53.9	1538	.121
Hematoma/contusion	11.1	290	13.6	385	.277
Fracture	26.2	682	15.8	450	.812
Other bone injury	5.8	151	3.4	96	.572
Meniscus/cartilage	7.1	185	5.1	146	.652
Tendon injury	12.6	329	4.9	140	.865
Injury mechanism					
Non-contact	128.3	3377	72.5	2099	.010*
Contact	94.9	2472	78.2	2166	.564
Recurrent	55.4	1445	18.2	519	.009*
Total	223.1	5815	150.8	4303	.026*

*Statistically significant difference between the Standard-11+ and $P2_{post}$ groups.

injuries/1000 h, total injuries = 171) to the 2016 baseline injury incidence data (IIR = 19.9 injuries/1000 h, total injuries = 558)[6] showed that both the Standard and $P2_{post}$ groups displayed reduced injury rates of 38% and 35%, respectively, to the 2016 injury incidence (IRR = 0.63; 95% CI–0.48-0.68; clinical inference—*very likely beneficial*; 99.3%).

4 | DISCUSSION

This is the first study to evaluate whether manipulating delivery of the 11+ program can enhance program effectiveness and compliance. Simply rescheduling, such that Parts 1 and 3 are performed at the beginning and Part 2 of the 11+ program at the end of training, reduced the severity and burden associated with the most common injuries observed in football, while increasing individual player 11+ dose. The specific effects of rescheduling are discussed below.

4.1 | Effect on program effectiveness

Irrespective of how the 11+ program was scheduled, the injury incidence rate was reduced (Standard-11+ = 38% reduction; $P2_{post}$ = 40% reduction compared with 2016 baseline) in this study. These reductions were consistent with previous research,[6,11] and our results show that the 11+ program is equally as effective in reducing injury incidence in football, whether performing all three parts collectively at the start of training or with Part 2 rescheduled until the end of training. There were, however, a significantly lower total number of severe injuries and days lost to injury observed in the $P2_{post}$ group in this study. Players in the $P2_{post}$ group performed the 11+ more frequently than the Standard-11+ group, which may have resulted in greater physiological adaptions to the 11+ in the $P2_{post}$ group. The 11+ has been shown to result in both acute and chronic performance benefits, including speed and agility, in addition to potential injury reduction effects, such as improving strength, balance, muscle activity, and core stability.[29,35,36] It is therefore plausible that the benefits observed in the $P2_{post}$ group are related to a dose effect, rather than any physiological changes in response to the scheduling of exercises.

Performing the exercises contained in Part 2 of the 11+ before training is the source of most concern for practitioners and is the most modified component of the 11+.[15] As such, investigating methods to improve compliance and effectiveness

TABLE 4 Muscle[a] and ligament injury pattern, incidence, and burden for participants in the Standard-11+ and $P2_{post}$ groups

	Standard 11+ group				$P2_{post}$ group						
	No. of injuries	IR per 1000 h [95% CI]	Days lost per 1000 h	Total days lost	No. of Injuries	IR per 1000 h [95% CI]	Days lost per 1000 h	Total days lost	IRR[b] [95% CI]	Clinical Inference[c] - IRR	*P* value (No. Injuries)	*P* value (Total days lost)
Non-contact muscle												
Thigh	56	2.3 [1.7,2.7]	22.1	1092	56	1.8 [1.4,2.2]	20.1	672	0.8 [0.5, 1.1]	Possibly beneficial	.395	.010*
Hamstrings	49	1.9 [1.3,2.5]	35.8	937	44	1.4 [1.0,2.0]	17.0	552	0.7 [0.5, 1.1]	Possibly beneficial	.285	.005*
Quadriceps	10	0.4 [0.1,0.9]	4.6	121	11	0.4 [0.1,1.1]	3.2	116	1.0 [0.4, 2.4]	Unclear	.999	.580
Hip/groin	34	1.2 [0.7,1.9]	15.3	456	38	1.3 [0.9,1.8]	13.6	404	1.1 [0.7, 1.8]	Possibly trivial	.678	.882
Lower leg	27	1.0 [0.6,1.5]	11.0	286	23	0.8 [0.4,1.4]	5.2	151	0.8 [0.4, 1.5]	Unclear	.556	.231
Non-contact ligament												
Ankle	45	1.7 [1.1,2.5]	25.3	659	31	1.1 [0.7, 1.7]	15.5	441	0.6 [0.3, 0.9]	Likely beneficial	.131	.449
Anterior Cruciate Ligament Rupture	8	0.15 [0.01, 0.4]	22.3	1280	3	0.06 [0.01, 0.2]	9.1	474	0.4 [0.1, 0.6]	Unclear	.238	.237

[a]Muscle injuries only included structural and functional injuries (ie, muscle injuries excluded contusions, hematoma, and tendon-related injuries).
[b]IRR—incidence rate ratio for injury incidence ($P2_{post}$ group: Standard-11+ group).
[c]Clinical inference determined from IRR via Hopkins "combine and compare" analysis.[34]
*Statistically significant difference between the Standard-11+ and $P2_{post}$ groups.

of exercises in Part 2 of the 11+ program such as the Nordic Hamstring Exercise (NHE), which is known to reduce the risk of hamstring injuries,[37] is important for improving adoption.[38] Previous research has suggested that the scheduling of the NHE has no impact upon chronic strength gains, albeit the muscle architectural mechanism seems to differ.[24] In the current study, however, the incidence of hamstring injury was similar in the Standard-11+ and $P2_{post}$ groups, and collectively 50% lower than our previous research,[6] suggesting that scheduling of the NHE does not impact effectiveness. Performing NHEs prior to training can, however, transiently reduce eccentric hamstring strength, which can in turn increase injury risk.[22] This can contribute to negative perceptions of the 11+ because fatigue and soreness from the NHE are reported barriers to 11+ program adoption.[10,16,17] Performing NHEs at the start of training in the Standard-11+ group did not increase training-related hamstring injury risk with most hamstring injuries occurring during matches. Interestingly, however, a significantly lower time lost, and subsequently severity, of hamstring injury was observed in the $P2_{post}$ group. This finding was in contrast to previous research in which the inclusion of the NHE was not associated with a reduction in hamstring injury severity.[25] Considering our finding and research that has shown hamstring eccentric strength decreases as a match progresses,[39] performing the NHE, as a component of a larger injury prevention program, after training might be an effective strategy for reducing hamstring injury incidence and burden.

A significantly lower incidence of ankle injuries was also observed in the $P2_{post}$ group relative to their Standard-11+ counterparts. Ankle sprains most commonly occur in the later stages of matches,[40] with ankle function changing under fatigue.[41] Exercises that focus on improving balance have been shown to be more effective when performed after football training[42] and therefore suggest that performing the ankle stability exercises in Part 2[43] at the end of training, in addition to the higher dose exposure, is likely to have improved the effectiveness of this component of the 11+ program. However, the $P2_{post}$ group incurred a significantly higher quadriceps injury incidence compared with the Standard-11+ group. The higher quadriceps injury incidence was a consequence of a significantly higher number of anterior thigh contusion injuries incurred by the $P2_{post}$ (23 vs 7 injuries), whereas non-contact quadriceps muscle strains were similar between groups. The 11+ program is designed to reduce non-contact injuries.[44] Therefore, the higher quadriceps injury incidence in the $P2_{post}$ group is unlikely the result of the program rescheduling.

Recurrent injuries are problematic in sub-elite football with inadequate recovery, poor physical conditioning on return to play, and a lack of access to medical care believed to contribute to the high incidence rate.[6,9,45] Interestingly, recurrent injury incidence and time lost to recurrent injury

were significantly lower in the $P2_{post}$ group compared with the Standard-11+, and both groups had a lower injury incidence when compared to the 2016 cohort.[6] Previous research has shown that increased compliance to strength programs is associated with reduced injury incidence.[12,19] Our results additionally suggest that rescheduling the exercises in Part 2 so that they are performed more regularly reduces ankle injury incidence, hamstring injury severity, and injury recurrence. We speculate that by significantly reducing the number of severe injuries and reducing time lost to injury, players in the $P2_{post}$ group returned to training earlier, increased their exposure to the 11+ program as well as to football training, and, in turn, reduced the injury risk caused by de-training for the most common injuries in football.[6,29,45-47]

To our knowledge, this study is the first to analyze the injury burden associated with the 11+ program, allowing for an examination beyond injury incidence.[33] A 33% lower injury burden was observed in the $P2_{post}$ group compared with both the Standard-11+ group and to the 2016 baseline,[6] with the greatest burden reductions associated with the most common injuries (ankle sprains, hamstring, and calf muscle strains) in football.[6,33] Lower time lost for hamstring, quadriceps, and calf injuries and lower injury incidence for ankle sprains account for the lower injury burden observed in the $P2_{post}$ group compared with the Standard-11+ group. Interestingly, the Standard-11+ showed a reduction in injury burden compared with the 2016 baseline,[6] which is likely to be the result of a significantly higher number of severe injuries in the baseline season. Additionally, injury burden associated with anterior cruciate ligament rupture (ACLR) in the $P2_{post}$ group was lower compared with both the Standard 11+ group and 2016 data,[6] with ACLR incidence 2.5 times lower in the $P2_{post}$ compared with the Standard-11+ group, and half that of 2016 baseline.[6] Furthermore, it is noteworthy that 6 of the 8 ACLR in the Standard-11+ group were non-contact injuries, while all ACLR in the $P2_{post}$ group involved contact with another player. Previous 11+ research found a similar dose-related effect to ACLR incidence,[48] and our findings present encouraging data that may help reduce non-contact ACLR in sub-elite football.

4.2 | The potential role of rescheduling on 11+ compliance

Rescheduling Part 2 of the 11+ to the end of training significantly increased program compliance with a 20% higher number of 11+ doses observed in the $P2_{post}$ compared with the Standard-11+ group. When considering the percentage of 11+ doses relative to training sessions completed, the $P2_{post}$ group individual player dose (57.7%) was higher than has previously been reported (47%) in 11+ research in youth football,[49] whereas the Standard-11+ group was lower (32.7%). However, compliance in both groups in the current study would be categorized as "low" and "moderate"[12] or "low"[50] relative to previous research. This result was despite "best practice" strategies to encourage program compliance, including extensive coach and staff education[50] and engagement with stakeholders.[26] The previous 11+ studies, however, did not record or state whether *all* components of the 11+ program were completed for an exposure to be recorded.[12,50] The apparently low compliance in this study might have been a consequence of the strict compliance criteria we applied, as only "doses" in which players completed *all* three components of the 11+ program were included in the analysis.

4.3 | Limitations

Performing large scale injury research across numerous clubs can result in several methodological limitations that must be acknowledged. We acknowledge potential issues that may arise from performing multiple sub-classification hypothesis testing on the same data set. Initial power calculations for subject inclusion were based on evaluating total injury incidence and burden. Once the overall effect was determined, we performed sub-classification analysis on different injury locations and types, which will reduce the power size of the sample analyzed. To overcome the potential impact of obtaining a false-positive or false-negative result, we determined clinical inferences[34] to allow for practical implications to be drawn from the findings. Although caution may need to be applied to the sub-class findings, the number of injuries recorded for specific injuries, such as hamstring muscle injuries, in our study was larger compared with other published research.[25,37]

Attempts were made to control the delivery of 11+ exercises for both groups to allow for the efficacy of the scheduling change to be correctly determined and compliance accurately assessed. A limitation to the application of the "compliance" assessment when evaluating the 11+ program is the progression of stages in Part 2. There is not a specific progression of Part 2 exercises within the program, and apart from the restrictions applied at the initial stages, standardized progression through these stages for players is not possible. We attempted to address this issue by only "allowing" players to progress through a stage in Part 2 once the PDC or coach was satisfied with the technique and performance of the exercise and, as such, maintained control over the prescription of exercises for the players. Although this limitation may be indicative of a pragmatic issue associated with real-world program coordination, we applied the more rigorous compliance definition to ensure the true effect of the rescheduling Part 2 was evaluated.

Notably, while a PDC was present to coordinate the 11+ program, the quality of how well the exercises were performed was not recorded. It is possible that how well the 11+ exercises were performed and extra exercises may have

impacted on injury incidence outcomes. It should also be acknowledged that the presence of the PDC at training sessions may have facilitated the compliance observed in this study.

Multiple PDCs were assigned across different clubs, which might have resulted in under- or overreporting of injuries.[6] Moreover, variations in coaching styles,[51] player fitness and physical characteristics, and previous injury history of players were not considered in the analysis.[52] We attempted to standardize the knowledge base, program implementation, and data collection by providing an extensive education program for coaches and PDCs before the intervention was implemented to minimize these limitations. Additionally, PDCs performed several practice injury reports before the season started to improve interclub data reporting consistency. Further research, however, is necessary to determine whether the improved 11+ outcomes in the $P2_{post}$ group were due to increased dose exposure or the scheduling change, or a combination of the two elements. The long-term physiological adaptions to performing Part 2 post-training also require further investigation. Finally, the translation of the results and conclusions from this study to other populations, such as youth and females, should be approached with caution. It is beyond the scope of this study and as such we do not speculate as to the impact of rescheduling Part 2 of the 11+ would have beyond the semi-professional men's cohort.

5 | CONCLUSION

Rescheduling exercises in Part 2 of the 11+ maintained program effectiveness while increasing compliance, thereby combatting some of the barriers associated with uptake of the 11+. Our results also suggest that improving 11+ compliance and performing Part 2 exercises at the end of training reduce the injury burden and severity associated with the most problematic injuries in football.

5.1 | Perspectives

Despite the success of the 11+ program in reducing injury incidence in football,[11] a number of barriers exist which limit implementation of the program.[15,16] This study shows that the simple act of rescheduling Part 2 of the 11+ program to the end of training significantly improves 11+ compliance while maintaining the overall effectiveness of the 11+ program. Additionally, rescheduling appeared to impact directly upon severe injury incidence, reducing the number of days lost to injury for the most common injuries in football. The findings of this study provide a simple and practical method to potentially improve the effectiveness of the 11+ program and the compliance to the 11+ program, and thus may assist in improving the overall adoption of the program.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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